Waquar Kaleem

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Education

- Ph.D. in Industrial Engineering and Operations Research, Penn State, 2022 – 2027 (May)
  Advisor: Dr. Anirudh Subramanyam
  Dissertation: *Scalable AI-integrated optimization for electrified transportation and logistics*
  Committee: Drs. Paul Griffin, Aron Laszka, S. Ilgin Guler, Changhyun Kwon
- M.S. in Industrial Engineering and Operations Research, Penn State, 2024 (en route to the Ph.D.)
- B.Tech. in Petroleum Engineering, Rajiv Gandhi Institute of Petroleum Technology (RGIPT), India, 2020
  Thesis: *Machine learning for lithology and hydrocarbon zone identification from well log data*
  Director's Gold Medal, awarded to one graduating student across all disciplines

Research Interests

Models, algorithms, and software for AI-integrated discrete optimization and sequential decision-making under uncertainty, using high-performance and GPU computing to improve decision-making in network design, supply-chain planning, freight and logistics operations, and critical transportation infrastructure.

Honors & Awards

- IISE Future Faculty Fellow, selected nationally, 2025–2026.
- Argonne Training Program on Extreme-Scale Computing (ATPESC), U.S. Department of Energy, 2026. Approximately 20% acceptance rate. **Newswise**
- Marcus Graduate Student Scholarship (Penn State), $10,000, one of six from twenty-two, 2025.
- Selected for the Deep Learning for Science Summer School (DL4Sci), Lawrence Berkeley National Laboratory, 2025. Approximately 35% acceptance rate.
- Honorable Mention, Poster Award (Penn State AI Hub), 2025. **Penn State News**
- Travel Award (Mixed Integer Programming Workshop), 2025.
- Best Paper (IISE Logistics and Supply Chain Division), 2024. **Award announcement**
- Oil and Natural Gas Corporation Scholarship (Government of India), 2021.
- S. N. Bose Scholar (Indo-US Science and Technology Forum), 2019. **Scholar blog**
- DAAD WISE Research Scholarship (Germany), 2019.
- AAPG L. Austin Weeks Undergraduate Research Grant, $500, 2019.
- Third Place, Best Paper (SPE South Asia Regional Student Paper Contest), 2019.

Proposals and Funded Projects

1. *Climate-Resilient Electric Bus Infrastructure Planning*, Lee Schipper Memorial Scholarship for Sustainable Transport and Energy Efficiency, Volvo Research and Educational Foundations, 2026. **Top-ten finalist globally** (Global Scholar track); not funded. Role: proposal preparation.
2. *Offline Model-Based Discrete Optimization: Theory, Algorithms, and Applications to Service Systems*, The Harold and Inge Marcus Endowment for Technion/PSU IE Partnership, $24,000, 2024–2026. PI: Dr. Anirudh Subramanyam. Role: assisted in internal document preparation.

Publications

Refereed Journal Articles

1. **W. Kaleem**, T. Cokyasar, J. Larson, O. Verbas, T. H. Bhuiyan, A. Subramanyam. Extreme-scale EV charging infrastructure planning for last-mile delivery using high-performance parallel computing. *Transportation Research Part B: Methodological*, 205, 103403, 2026. DOI: 10.1016/j.trb.2026.103403
2. S. H. Hosseini Dolatabadi, T. H. Bhuiyan, **W. Kaleem**, A. Subramanyam, T. Cokyasar. A branch-and-cut algorithm for routing a heterogeneous drone-ground vehicle fleet to deliver time-sensitive products. *Transportation Research Part C: Emerging Technologies*, 183, 105454, 2026. DOI: 10.1016/j.trc.2025.105454
3. **W. Kaleem**, S. Tewari, M. Fogat, D. A. Martyushev. A hybrid machine learning approach based study of production forecasting and factors influencing the multiphase flow through surface chokes. *Petroleum*, 10(2), 354–371, 2024. DOI: 10.1016/j.petlm.2023.06.001. **Among the journal's most cited articles since 2023**.
4. **W. Kaleem**, K. A. Khan, A. Saxena. Statistical feature engineering using discrete wavelet transform for improving the prediction of compressional and shear sonic travel time logs. *Petroleum Science and Technology*, 43(3), 219–240, 2023. DOI: 10.1080/10916466.2023.2292781
5. K. R. Chaturvedi, D. Ravilla, **W. Kaleem**, P. Jadhawar, T. Sharma. Impact of low salinity water injection on CO_2 storage and oil recovery for improved CO_2 utilization. *Chemical Engineering Science*, 229, 116127, 2021. DOI: 10.1016/j.ces.2020.116127
6. **W. Kaleem**, A. Kumar, D. Panda. Luminescent N, S-doped carbon nanodot: an effective two-fluorophore system of pyridone and thiazolopyridone. *The Journal of Physical Chemistry C*, 122(46), 26722–26732, 2018. DOI: 10.1021/acs.jpcc.8b08249

Refereed Conference Proceedings

7. **W. Kaleem**, D. Lee, C. Kwon, A. Subramanyam. Neural embedded mixed-integer optimization for location-routing problems. *NeurIPS Workshop on Differentiable Learning of Combinatorial Algorithms*, 2025. URL: openreview.net/forum?id=WCILzWMXq9
8. **W. Kaleem**, H. Ayala, A. Subramanyam. Neural embedded optimization for integrated location and routing problems. *IISE Annual Conference Proceedings*, 1–6, 2024. DOI: 10.21872/2024iise_6627. **Best Paper Award, IISE Logistics and Supply Chain Division**. **Award announcement**
9. S. Tewari, P. Jain, S. Sahu, **W. Kaleem**, P. K. Jain. A novel surface roughness estimation and optimization model for turning process using the RSM-JAYA method. *Data Engineering and Applications, IDEA 2022*, Lecture Notes in Electrical Engineering, vol. 1146, 137–153, Springer, Singapore, 2024. DOI: 10.1007/978-981-97-0037-0_11

Under Review and Available Manuscripts

10. **W. Kaleem**, D. Lee, C. Kwon, A. Subramanyam. Neural embedded mixed-integer optimization for location-routing problems. Major revision (first round). PREPRINT: arXiv:2412.05665
11. **W. Kaleem**, T. Cokyasar, A. Ismael, O. Sahin, H. Uhm. Approximate dynamic programming for matching, scheduling, and pricing in truckload freight markets. Accepted for presentation at the *106th Transportation Research Board Annual Meeting*, 2027.
12. **W. Kaleem**, A. Subramanyam. Permutation invariant neural estimators for vehicle routing costs.

Software

1. **W. Kaleem**, D. Lee, C. Kwon, A. Subramanyam. NEO-LRP. Neural embedded optimization for location-routing problems (written in Python). URL: `github.com/Subramanyam-Lab/NEO-LRP`
2. Contributor, with accepted pull requests, to two open-source optimization packages. cuOpt.jl, a Julia interface to the NVIDIA cuOpt solver. URL: `github.com/jump-dev/cuOpt.jl` VRP-SolverEasy, a branch-cut-and-price exact solver for vehicle routing problems. URL: `github.com/inria-UFF/VRPSolverEasy`

Working Papers (In Preparation)

1. **W. Kaleem**, A. Subramanyam, A. U. Raghunathan. Tractable EV Depot Scheduling with Closed-Form Nonlinear Charging Dynamics and Physical Vehicle Movement.
2. **W. Kaleem**, E. Lopez Ledezma, A. Subramanyam. Neural embedded mixed-integer optimization for structurally decomposable vehicle routing problems.
3. **W. Kaleem**, T. Kwon, H. Lefebvre, A. Subramanyam. A survey on optimization with feedforward neural network constraints: models, methods, and applications.
4. **W. Kaleem**, A. M. Kazachkov, A. Subramanyam. An embarrassingly parallel probe, dive, and propagate heuristic on GPU for mixed-integer programming.

Patents

1. **W. Kaleem**, A. U. Raghunathan. Systems and methods for multi-timescale scheduling and control of energy charging infrastructure. Filed with the United States Patent and Trademark Office, 2026. Joint filing with *Mitsubishi Electric Research Laboratories*.

Research Experience

Argonne National Laboratory, Lemont, IL. May – August 2026

Research Aide Technical, Transportation and Power Systems Division.
Supervisor: Dr. Taner Cokyasar.
Robust EV charging infrastructure planning under decision-dependent uncertainty.
Value function approximation for matching and pricing in truckload freight markets.

Mitsubishi Electric Research Laboratories, Cambridge, MA. September – December 2025

Optimization Research Intern. Supervisor: Dr. Arvind U. Raghunathan.
Grid-connected EV charging schedule optimization, filed as a joint patent.

Argonne National Laboratory, Lemont, IL. May – August 2024

Givens Associate, Mathematics and Computer Science.
Supervisors: Drs. Jeffrey M. Larson and Taner Cokyasar.
Stochastic charger location and allocation for last-mile delivery.
Lagrangian decomposition with cutting planes across 680 CPU cores. The method solved the largest instance reported in the literature.

Rezlytix Technologies, Remote. July – October 2020

Machine Learning Research Intern.
Synthetic seismic datasets and machine learning models that reached 87 percent accuracy, above conventional three-dimensional mapping.

The Pennsylvania State University, University Park, PA. May – August 2019

Undergraduate Research Assistant. Advisor: Dr. Sanjay Srinivasan.
Lattice Boltzmann simulator for modeling CO_2 plume migration in saline reservoirs.
S. N. Bose Scholar, Indo-US Science and Technology Forum. **Scholar blog**

Teaching Experience

Learning Assistant, Penn State Summer 2025

CMPEN/EE 454, Computer Vision. Instructor: Dr. Mohamed Almekkawy.
Guided students through code debugging in weekly problem sessions, held office hours, graded homework assignments and exams, and corrected errors in the course materials.
Student feedback: "*Thank you for all of your help and guidance in EE 454. You were very kind and I appreciate all of your guidance!*"

Graduate Teaching Assistant, Penn State Spring 2025

IE 402, Engineering Economy. Instructor: Dr. Qiushi Chen.
Graded assignments, quizzes, and exams for more than thirty students, held weekly office hours, and helped prepare course materials.

Lab Instructor and Teaching Assistant, Penn State Fall 2024

IE 418, Advanced Human-Computer Interaction. Instructor: Dr. Hari Thiruvengada.
Led two weekly lab sections on AI interaction design, chatbot building, and workflow automation for more than sixty students. Opened each session with a walkthrough of the tasks and drew on research and industry practice to connect lab exercises to real applications. Rotated randomized teams to broaden peer interaction. Held office hours and graded assignments and exams.
Student feedback: "*Waqar has been the most helpful especially during lab in understanding and implementing the material.*"

Undergraduate Teaching Assistant, RGIPT Spring 2020

MTH 210A, Numerical and Statistical Methods.

Teaching Development

Schreyer Conference on Alternative Grading, Schreyer Institute for Teaching Excellence, Penn State March 2026

ENGR 888, Seminar for Engineering Teaching Assistants, Leonhard Center Fall 2024

Graduate seminar that covers the alignment of course objectives with instruction and assessment, methods for active and collaborative learning, classroom assessment design, and academic integrity practice.
Mid-semester feedback study: collected feedback from seventeen IE 418 students and adjusted my teaching to make the lab content more accessible.
Peer observation: observed an IE 327 lab, Introduction to Work Design, and was observed teaching an IE 418 lab.

Advising and Mentorship

- Eduardo Lopez Ledezma (Penn State) 2023 – 2026

  Undergraduate researcher.
  Topic: Neural embedded optimization for vehicle routing problems.
  Working paper: *Neural embedded mixed-integer optimization for structurally decomposable vehicle routing problems*.
  Placement: FTR Trader, Engelhart Commodities Trading Partners.

- Rudradip Khanra (RGIPT) 2025 – 2026

  Undergraduate researcher.
  Topic: Machine learning methods for production forecasting.
  Acknowledged in preprint: *A simple gated recurrent network for multi-flow production forecasting*.

Industry Experience

Oil and Natural Gas Corporation, Rajahmundry, India. November 2020 – July 2022

Assistant Executive Engineer (Drilling).
Assisted onshore drilling operations, reviewing geological technical orders and coordinating rig preparation, equipment movement, and blowout preventer (BOP) assembly. Later supported the Mine Safety Officer with regulatory compliance across more than five rigs.

Talks

Conference Talks

Neural embedded mixed-integer optimization for location-routing problems
- INFORMS TSL Conference, MIT, Cambridge, MA July 2026
- INFORMS Computing Society Conference, Toronto, Canada March 2025

Optimization over deep ReLU networks using penalty methods
- INFORMS Annual Meeting, Atlanta, GA October 2025

Extreme-scale EV charging infrastructure planning for last-mile delivery
- INFORMS Computing Society Conference, Toronto, Canada March 2025

Neural embedded optimization for sets
- IISE Annual Conference, Montreal, Canada May 2024
- INFORMS Optimization Society Conference, Houston, TX March 2024

Neural embedded optimization for integrated location and routing problems

IISE Annual Conference, Montreal, Canada — May 2024

Seminars and Symposia

Approximate dynamic programming for matching, scheduling, and pricing in truckload freight markets

Transportation and Mobility Group, Argonne National Laboratory — August 2026

Extreme-scale EV charging infrastructure planning for last-mile delivery

Summer Argonne Student Symposium (SASSy), Lemont, IL — August 2024

Poster Presentations

Extreme-scale EV charging infrastructure planning

POLARIS User Group (PUG) Workshop, Chicago, IL (invited) — June 2026

Mixed Integer Programming Workshop, Minneapolis, MN — June 2025

Offline model-based routing optimization with MIP-representable deep set surrogates

Mixed Integer Programming Workshop, Stamford, CT — May 2026

Neural embedded mixed-integer optimization for location-routing problems

AI Week, Penn State. **Penn State News** — April 2025

AAAI Bridge Program, Philadelphia, PA — February 2025

Large-scale stochastic charger location and allocation in last-mile delivery

Transportation Research Board Annual Meeting, Washington, DC — January 2025

Neural embedded optimization for sets

Symposium on Computational Mathematics, Penn State — November 2023

ICDS Symposium, Penn State — October 2023

Professional Service

Referee for Peer-Reviewed Journals and Conferences

Transportation Research Part C, European Journal of Operational Research, Engineering Applications of Artificial Intelligence, Computers and Industrial Engineering, Geoscience and Energy, Transportation Research Board Annual Meeting, IISE Annual Conference, AAAI/ACM Conference on AI, Ethics and Society, SBP-BRiMS.

Leadership and Committees

- Secretary, INFORMS Student Chapter, Penn State, 2024–2026.
- Committee Member, SPE Annual Technical Conference and Exhibition, Data Science Engineering and Analytics Subcommittee, 2022–2024.
- Volunteer organizer, International Conference on Unconventional Energy Resources (ICUER), 2019. **Best Volunteer Award**.
- President, Society of Exploration Geophysicists Student Chapter, RGIPT, 2020.
- Organizing Secretary, Science and Technical Committee, RGIPT, 2019.

Outreach

- Volunteer, Gyan-Arpan, a student initiative that taught children from local schools, RGIPT, 2017–2018.

Technical Skills

Operations Research: Linear Programming, Mixed-Integer Programming, Nonlinear Programming, Stochastic Programming, Simulation Optimization; Lagrangian decomposition, column generation, branch-and-price, and branch-and-cut; metaheuristics, matheuristics, and primal heuristics

Languages and Tools: Python, SQL, Julia, Bash, MATLAB, Git, LaTeX; Gurobi, CPLEX, Pyomo, HiGHS, OR Tools, CuOpt, PyTorch, Scikit-Learn, DeepHyper

High Performance Computing: MPICH, MPI4py, Slurm, CUDA, OpenMP, Singularity, PETSc/TAO, TAU; parallel and distributed programming

Compute Platforms: Penn State ROAR (680 CPU cores across heterogeneous partitions; A100 and P100 GPUs), University of Florida HiPerGator (H100 and B200 GPUs), NVIDIA Brev cloud GPUs (L40S, H100, and A100), NERSC Perlmutter (Lawrence Berkeley National Laboratory), and the exascale machines Aurora (Argonne National Laboratory) and Frontier (Oak Ridge National Laboratory)

Machine Learning: Large-scale hyperparameter optimization, model validation, neural network surrogates

Quantum Computing: Exposure through ATPESC, running circuits on the IQM Resonance cloud platform with the 20-qubit Garnet and the 54-qubit Emerald superconducting processors

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS); Institute of Industrial and Systems Engineers (IISE); Mathematical Optimization Society (MOS); Society of Petroleum Engineers (SPE)